

FANUC Robot **series**

R-30*i*B and R-30*i*B PLUS CONTROLLER

**MILLER WELD EQUIPMENT
LIBRARY OPTION**

SETUP AND OPERATIONS MANUAL

Version 8.10 and later

MAROBMWLD02131E REV. D

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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Patents

One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

4,630,567 4,639,878 4,707,647 4,708,175 4,708,580 4,942,539 4,984,745
5,238,029 5,239,739 5,272,805 5,293,107 5,293,911 5,331,264 5,367,944
5,373,221 5,421,218 5,434,489 5,644,898 5,670,202 5,696,687 5,737,218
5,823,389 5,853,027 5,887,800 5,941,679 5,959,425 5,987,726 6,059,092
6,064,168 6,070,109 6,086,294 6,122,062 6,147,323 6,204,620 6,243,621
6,253,799 6,285,920 6,313,595 6,325,302 6,345,818 6,356,807 6,360,143
6,378,190 6,385,508 6,425,177 6,477,913 6,490,369 6,518,980 6,540,104
6,541,757 6,560,513 6,569,258 6,612,449 6,703,079 6,705,361 6,726,773
6,768,078 6,845,295 6,945,483 7,149,606 7,149,606 7,211,978 7,266,422
7,399,363

FANUC CORPORATION Patent List

4,571,694 4,626,756 4,700,118 4,706,001 4,728,872 4,732,526 4,742,207
4,835,362 4,894,596 4,899,095 4,920,248 4,931,617 4,934,504 4,956,594
4,967,125 4,969,109 4,970,370 4,970,448 4,979,127 5,004,968 5,006,035
5,008,834 5,063,281 5,066,847 5,066,902 5,093,552 5,107,716 5,111,019
5,130,515 5,136,223 5,151,608 5,170,109 5,189,351 5,267,483 5,274,360
5,292,066 5,300,868 5,304,906 5,313,563 5,319,443 5,325,467 5,327,057
5,329,469 5,333,242 5,337,148 5,371,452 5,375,480 5,418,441 5,432,316
5,440,213 5,442,155 5,444,612 5,449,875 5,451,850 5,461,478 5,463,297
5,467,003 5,471,312 5,479,078 5,485,389 5,485,552 5,486,679 5,489,758
5,493,192 5,504,766 5,511,007 5,520,062 5,528,013 5,532,924 5,548,194
5,552,687 5,558,196 5,561,742 5,570,187 5,570,190 5,572,103 5,581,167
5,582,750 5,587,635 5,600,759 5,608,299 5,608,618 5,624,588 5,630,955
5,637,969 5,639,204 5,641,415 5,650,078 5,658,121 5,668,628 5,687,295
5,691,615 5,698,121 5,708,342 5,715,375 5,719,479 5,727,132 5,742,138
5,742,144 5,748,854 5,749,058 5,760,560 5,773,950 5,783,922 5,799,135
5,812,408 5,841,257 5,845,053 5,872,894 5,887,122 5,911,892 5,912,540
5,920,678 5,937,143 5,980,082 5,983,744 5,987,591 5,988,850 6,023,044
6,032,086 6,040,554 6,059,169 6,088,628 6,097,169 6,114,824 6,124,693
6,140,788 6,141,863 6,157,155 6,160,324 6,163,124 6,177,650 6,180,898
6,181,096 6,188,194 6,208,105 6,212,444 6,219,583 6,226,181 6,236,011
6,236,896 6,250,174 6,278,902 6,279,413 6,285,921 6,298,283 6,321,139
6,324,443 6,328,523 6,330,493 6,340,875 6,356,671 6,377,869 6,382,012
6,384,371 6,396,030 6,414,711 6,424,883 6,431,018 6,434,448 6,445,979
6,459,958 6,463,358 6,484,067 6,486,629 6,507,165 6,654,666 6,665,588
6,680,461 6,696,810 6,728,417 6,763,284 6,772,493 6,845,296 6,853,881
6,888,089 6,898,486 6,917,837 6,928,337 6,965,091 6,970,802 7,038,165
7,069,808 7,084,900 7,092,791 7,133,747 7,143,100 7,149,602 7,131,848
7,161,321 7,171,041 7,174,234 7,173,213 7,177,722 7,177,439 7,181,294
7,181,313 7,280,687 7,283,661 7,291,806 7,299,713 7,315,650 7,324,873
7,328,083 7,330,777 7,333,879 7,355,725 7,359,817 7,373,220 7,376,488
7,386,367 7,464,623 7,447,615 7,445,260 7,474,939 7,486,816 7,495,192
7,501,778 7,502,504 7,508,155 7,512,459 7,525,273 7,526,121

Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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1 OVERVIEW

1 OVERVIEW

The Miller Weld Equipment Library Option (R580) allows you to configure ArcTool to communicate with Miller arc welding power supplies that have different hardware and software interfaces. Currently, these four interfaces are supported:

1. **Deltaweld 452** (Requires a process I/O board)
2. **Auto-Invision** (Requires a process I/O board)
3. **Auto-Axcess DNET** (Requires DeviceNet hardware and the DeviceNet software option)
4. **Auto-Axcess ENET** (Requires the Ethernet I/O Scanner option R540)

The **Ethernet** interface was introduced in Version V8.10 (R-30iB controller only). FANUC America recommends using the latest release version available for this equipment.

The **Ethernet** interface is only available for use with the Weld Procedure programming technique which utilizes Weld Start and Weld End instructions. This is the default method of weld programming in V8.10. Note that Multipass programming in the TP program header is not supported with the Weld Procedures programming method.

The **DeviceNet** interface is not supported with weld procedures. It is only supported with the weld schedule programming technique which utilizes Arc Start and Arc End instructions. The DeviceNet interface can be used on R-30iA or R-30iB controllers.

2

DEVICENET

2 DEVICENET

2.1 Preliminary Steps

Before using the ArcTool setup menu at controlled start to select the Miller Auto-Axcess DNET interface you should perform these preliminary steps:

1. Plan the range of I/O port numbers used by the welder and other devices in the system.
2. Configure ArcTool Features (Multi-Process, Multi-Equipment and Weld Procedures)
3. Install the DeviceNet Hardware

These preliminary steps are described in the next few sections. After these steps are completed the Miller Auto-Axcess DNET interface can be selected in the controlled start menu as described in Procedure 1-2.

2.1.1 Plan the I/O range

If your robot system has other I/O devices it may be important to you where the welding I/O signals will be assigned within the overall I/O map.

The Miller DNET interface uses 64 digital inputs and 64 digital outputs. The first 16 inputs and outputs are individual control signals. The remaining 48 signals are configured as 6 group outputs and 6 group inputs.

During the controlled start setup procedure you have the opportunity to enter a start point for the Digital I/O signals and for the Group I/O signals. You can also simply allow the signal numbering to begin at the first available area.

Start points begin one after a multiple of 8 (1, 9, 17, etc.) The number you enter will be used to search for an available I/O range. The full range of contiguous signals (64 inputs and 64 outputs) must be available without conflict with a previous assignment.

2.1.2 Configure ArcTool Features

There are a few important ArcTool settings which must be configured at controlled start before selecting the Miller DNET interface.

Procedure 2-1 Configure ArcTool Features

Step

- 1 Perform a controlled start.
 - a If the controller is on, turn it off.
 - b On the teach pendant, press and hold the PREV and NEXT keys.
 - c While still pressing PREV and NEXT on the teach pendant, turn the power disconnect circuit breaker to ON. You will see a screen similar to the following.

```

----- CONFIGURATION MENU -----
-
1 Hot start
2 Cold start
3 Controlled start
4 Maintenance
  
```

- d After the configuration menu appears on the screen, release the PREV and NEXT keys.
 - e Select 3, Controlled start.
- 2 In the ArcTool Setup menu cursor to the **Multi-process** field and select DISABLED. The Miller DNET interface does not require this feature.
- 3 If you are using multiple weld equipment, cursor to the **Number of weld equipment** field and enter the correct number. This field cannot be set if the multi-equipment option is not installed.

```

ArcTool Setup

 8 Multi-process:          DISABLED
 9 Weld ID:                DISABLED
10 Number of weld schedules: 32
11 Number of weld equipment: 1

[ TYPE ]                  HELP
  
```

- 4 Beginning with version 8.10 ArcTool uses Weld Procedures as the default programming method. Weld Procedures use Weld Start and Weld End instructions instead of Arc Start and Arc End. This feature must be disabled to use the Miller **DeviceNet** interface. Weld Procedures are used with the Miller **Ethernet** interface.

- a Press the MENU key and select Variables.
 - b Cursor to the system variable \$AWSCFG and press ENTER.
 - c Cursor to the field \$WLD_PRC_ENA and set it to FALSE.
- 5 Turn the controller off.

2.1.3 Install the DeviceNet Hardware

Required hardware:

1. FANUC R-30iB Robot Controller
2. Miller Auto-Axcess DI Welding Power Supply
3. DeviceNet Interface Board
4. DeviceNet Cable
5. 120 Ohm Resistor

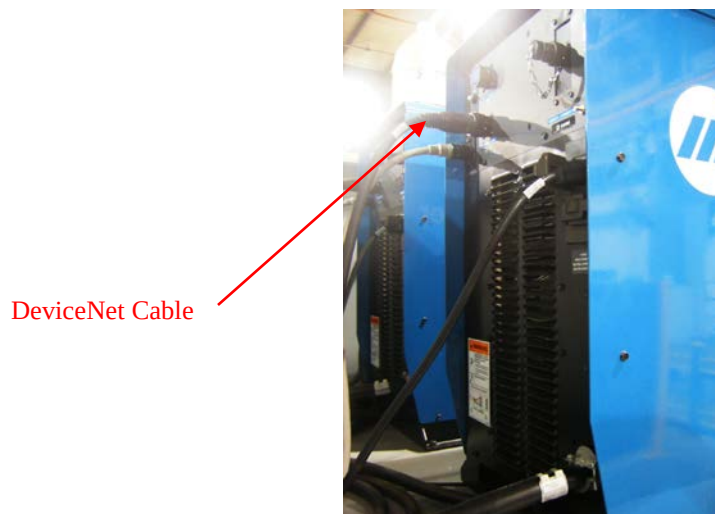
Procedure 2-2 Install DeviceNet Hardware

Conditions

- The robot controller is off.
- The Miller Power Supply is off.

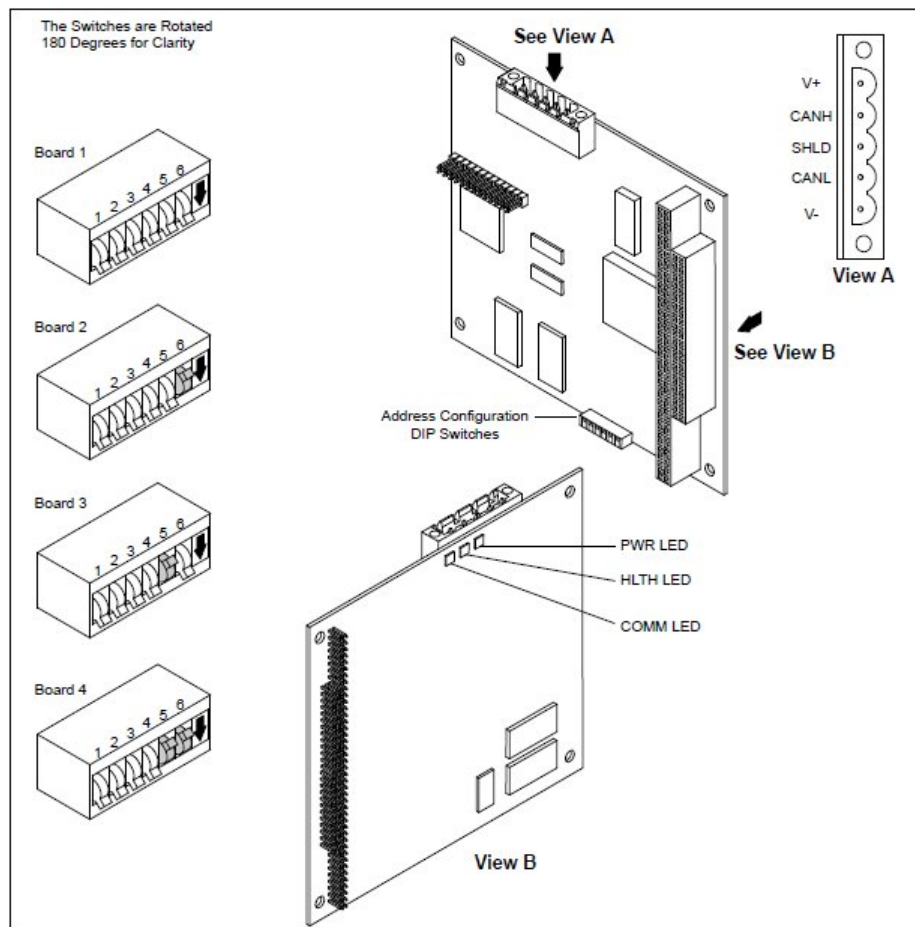
Step

- 1 Connect the DeviceNet cable to the connector located on the rear of the Miller welder.



- 2 Route the DeviceNet cable into robot controller cabinet.
- 3 Remove the connector from the end of the cable and strip back the insulation to expose the wires.
- 4 Insert the wires into the appropriate color coded DeviceNet cable plug. The drain wire must be in the center terminal. Do not allow drain wire to make contact with resistor.
- 5 Insert the 120 Ohm resistor into the blue and white terminals.
- 6 Set the dip switches on the DeviceNet board. The typical configuration is: All switches in the OFF position = Board 1 rack 81. See the *DeviceNet Interface Setup and Operations Manual* for more details about hardware configuration.
- 7 Install the DeviceNet board in the robot controller.

Note: The DeviceNet board does NOT need an external 24V supply from the robot.



2.2 Weld Power Supply Selection and Verification

Use the following procedure to configure ArcTool for operation with a Miller Auto-Axcess DNET weld power supply.

Procedure 2-3 Selecting a Miller Weld Power Supply at Controlled Start

Conditions

- The DeviceNet option is installed.
- The Miller Weld Equipment Library Option R580 is installed.
- You have completed the Preliminary Steps in section 2.1
- The ArcTool Setup menu is displayed at the Controlled Start as shown below.

ArcTool Setup		6/10
1 F Number:	F00000	
2 Welding setup:	USA	
3 Wire speed units:	IPM	
4 Weld speed units:	IPM	
5 Weld speed:	50	
6 Manufacturer:	General Purpose	
7 Model:	MIG (Volts, WFS)	
Press FCTN then START (COLD) when done.		
[TYPE]	CHECK	[CHOICE] HELP

Step

- 1 Move the cursor to the Manufacturer field and press F4, [CHOICE]. Select Miller. The Model name will default to Deltaweld452.
- 2 Move the cursor the Model field and press F4, [CHOICE] and select Auto-Axcess DNET from the list.
- 3 You will asked if you would like to specify a starting digital I/O number. If you choose not to specify a number, the I/O will be located at the first block of adequate size. If you specify a starting number the I/O will begin at or after this point. You will be asked the same question regarding the starting number for the group I/O.
- 4 Press FCTN, Select START(Cold), and press ENTER to Cold Start the controller.

2.2.1 Verify the DeviceNet Board Settings

Procedure 2-4 Checking the DeviceNet setup

Step

- 1 Press the MENU key and select I/O. Press F1, TYPE and select DeviceNet. You should see a screen similar to the one below.

I/O DeviceNet				
Board List				
			1/4	
Board	Comment		Rack	Status
1	[	]	81	<ONLINE >
2	[	]	82	OFFLINE
3	[	]	83	OFFLINE
4	[	]	84	OFFLINE
[TYPE] DEV-LST DEF-DEV DETAIL HELP >				

- 2 Press F4, DETAIL to verify the MAC ID setting.

I/O DeviceNet			
Board Detail			1/7
Board: 1		Status: ONLINE	
Scanner type: SST 5136-DN3-104			
Motherboard: Wide-mini			
1	MAC-Id:	27	
2	Baud-rate:	125 KB	
3	Board auto-restart:	ON	
4	Input resume state (rack 81): ZERO		
Slave Operation:			
Slave Status:		ONLINE	
5	Slave Error Severity:	WARN	
6	Size of output from master:	8 bytes	
7	Size of input to master:	8 bytes	
[TYPE]	LIST	HELP	

2.2.2 Verify the Digital I/O Assignments

The digital I/O signals were automatically configured during the controlled start setup. Comments were added to the 16 digital control signals, but not to those used as group I/O. Some of the signals in the Miller I/O range are not used.

I/O Digital Out						I/O Digital In					
#	SIM	STATUS	1/512			#	SIM	STATUS	1/512		
DO[1]	U	OFF	[E1	Reserved	]	DI[1]	U	OFF	[E1	Arc detect	]
DO[2]	U	OFF	[E1	Inch bwd	]	DI[2]	U	OFF	[E1	Bwd status	]
DO[3]	U	OFF	[E1	Inch fwd	]	DI[3]	U	OFF	[E1	Fwd status	]
DO[4]	U	OFF	[E1	Gas start	]	DI[4]	U	OFF	[E1	Gas status	]
DO[5]	U	OFF	[E1	Reserved	]	DI[5]	U	OFF	[E1	Touch fdbk	]
DO[6]	U	OFF	[E1	Reserved	]	DI[6]	U	OFF	[E1	Reserved	]
DO[7]	U	OFF	[E1	Reserved	]	DI[7]	U	OFF	[E1	Reserved	]
DO[8]	U	OFF	[E1	Reserved	]	DI[8]	U	OFF	[E1	Reserved	]
DO[9]	U	OFF	[E1	Reserved	]	DI[9]	U	OFF	[E1	Error	]
DO[10]	U	OFF	[E1	Weld start	]	DI[10]	U	OFF	[E1	Contactor	]
DO[11]	U	OFF	[E1	Reserved	]	DI[11]	U	OFF	[E1	Proc active	]
DO[12]	U	OFF	[E1	Touch cmd	]	DI[12]	U	OFF	[E1	Touch active	]
DO[13]	U	OFF	[E1	Program 0	]	DI[13]	U	OFF	[E1	Reserved	]
DO[14]	U	OFF	[E1	Program 1	]	DI[14]	U	OFF	[E1	Reserved	]
DO[15]	U	OFF	[E1	Program 2	]	DI[15]	U	OFF	[E1	Reserved	]
DO[16]	U	OFF	[E1	Reserved	]	DI[16]	U	OFF	[E1	Synergic	]
DO[17]	U	OFF	[		]	DI[17]	U	OFF	[		]
DO[18]	U	OFF	[		]	DI[18]	U	OFF	[		]
DO[19]	U	OFF	[		]	DI[19]	U	OFF	[		]
DO[20]	U	OFF	[		]	DI[20]	U	OFF	[		]
DO[21]	U	OFF	[		]	DI[21]	U	OFF	[		]
Sorted by port number.						Sorted by port number.					
[TYPE]	CONFIG	IN/OUT	ON	OFF	>	[TYPE]	CONFIG	IN/OUT	ON	OFF	>

Pressing the SHIFT-Wire +/- hard keys feeds the filler wire. This can be used to verify the digital I/O operation.

Pressing the CONFIG function key in the digital I/O screens confirms that there are 64 digital inputs and 64 digital outputs in the DeviceNet interface.

I/O Digital Out						I/O Digital In					
#	RANGE	RACK	SLOT	START	STAT.	#	RANGE	RACK	SLOT	START	STAT.
1	DO[1- 64]	81	27	1	ACTIV	1	DI[1- 64]	81	27	1	ACTIV
2	DO[65- 512]	0	0	0	UNASG	2	DI[65- 512]	0	0	0	UNASG

2.2.3 Verify the Group I/O Assignments

The group inputs and outputs signal names should match those shown in the following figure. The port numbers shown below are based on a starting port number of 1.

I/O Group Out					I/O Group In				
	#	SIM	VALUE	1/100		#	SIM	VALUE	1/100
GO[	1]	U	0	[E1 Weld Cmnd 1]	GI[	1]	U	0	[E1 Weld Fdbk 1]
GO[	2]	U	0	[E1 Weld Cmnd 2]	GI[	2]	U	0	[E1 Weld Fdbk 2]
GO[	3]	U	1	[E1 Req Param 1]	GI[	3]	U	1	[E1 Req Param 1]
GO[	4]	U	0	[E1 Req Param 2]	GI[	4]	U	0	[E1 Req Param 2]
GO[	5]	U	4	[E1 Sent Param 1]	GI[	5]	U	4	[E1 Sent Param 1]
GO[	6]	U	5	[E1 Sent Param 2]	GI[	6]	U	5	[E1 Sent Param 2]
GO[	7]	*	*	[	GI[	7]	*	*	[
GO[	8]	*	*	[	GI[	8]	*	*	[
Sorted by port number.					Sorted by port number.				
[TYPE] CONFIG IN/OUT SIMULATE UNSIM >					[TYPE] CONFIG IN/OUT SIMULATE UNSIM >				

The CONFIG function key can be used to show how many signals (NUM PTS) are mapped to each port.

I/O Group Out						I/O Group In					
GO #	RACK	SLOT	START PT	NUM	PTS	GI #	RACK	SLOT	START PT	NUM	PTS
1	81	27	17	16		1	81	27	17	16	
2	81	27	33	16		2	81	27	33	16	
3	81	27	49	4		3	81	27	49	4	
4	81	27	53	4		4	81	27	53	4	
5	81	27	57	4		5	81	27	57	4	
6	81	27	61	4		6	81	27	61	4	
7	0	0	0	0		7	0	0	0	0	
8	0	0	0	0		8	0	0	0	0	
[TYPE] MONITOR IN/OUT						[TYPE] MONITOR IN/OUT					
HELP >						HELP >					

2.2.4 Verify the Weld I/O Assignments

The weld input and output screens are shown below. In the DeviceNet interface the command parameters are always named Voltage and Wire feed. This is correct for the MIG welding mode. In the Pulse mode the first parameter is actually Arc Length.

I/O Weld Out					I/O Weld In				
				1/10					1/9
WELD SIGNAL	TYPE	#	SIM	STATUS	WELD SIGNAL	TYPE	#	SIM	STATUS
1 [Voltage	] GO[	1]	U	0.0	1 [Voltage	] GI[	1]	U	0.5
2 [Wire feed	] GO[	2]	U	0.0	2 [Current	] GI[	2]	U	0.0
3 [Weld start	] DO[	10]	U	OFF	3 [Arc detect	] DI[	1]	U	OFF
4 [Gas start	] DO[	4]	U	OFF	4 [Power fault	] DI[	9]	U	OFF
5 [Inch forward	] DO[	3]	U	OFF	5 [Wirestick	] DI[	5]	U	OFF
6 [Inch backward	] DO[	2]	U	OFF	6 [Gas fault	] [****]	*	***	
7 [Proc select 1	] DO[	13]	U	OFF	7 [Wire fault	] [****]	*	***	
8 [Proc select 2	] DO[	14]	U	OFF	8 [Water fault	] [****]	*	***	
9 [Proc select 3	] DO[	15]	U	OFF	9 [Arc enable	] [****]	*	***	
10 [Wire stick alarm]	[****]	*	***						
[TYPE]	HELP	IN/OUT	SIMULATE	UNSIM >	[TYPE]	HELP	IN/OUT	SIMULATE	UNSIM >

2.2.5 Verify the Weld Schedule Menu is Displayed

During the preliminary setup steps the system variable \$awscfg.\$wld_prc_ena was set to FALSE to disable Weld Procedures. If this step was completed, the first screen listed in the DATA menu should be Weld Schedules.

DATA Weld Sched				1/32
			IPM	COMMENT
1	20.0	200.0	20	
2	20.0	200.0	20	
3	20.0	200.0	20	
4	20.0	200.0	20	
5	20.0	200.0	20	
6	20.0	200.0	20	
7	20.0	200.0	20	
8	20.0	200.0	20	
9	20.0	200.0	20	
10	20.0	200.0	20	
Voltage	20.0	Volts (	0.0,	100.0)
[TYPE]	DETAIL			HELP >

2.3 Weld Process Menu Operation

In the SETUP menu, the Weld Process menu should be available in the [TYPE] menu.

SETUP Weld Proc			
			1/8
Selected Process: 1 [Process 1]			
1	Auto-Axcess	DNET Process 1	0
2	Auto-Axcess	DNET Process 2	1
3	Auto-Axcess	DNET Process 3	2
4	Auto-Axcess	DNET Process 4	3
5	Auto-Axcess	DNET Process 5	4
6	Auto-Axcess	DNET Process 6	5
7	Auto-Axcess	DNET Process 7	6
8	Auto-Axcess	DNET Process 8	7
[TYPE] SELECT HELP			

This menu is a list of the 8 weld processes that can be selected. You may cursor up and down to a weld process and use function key F3, SELECT, to manually select a weld process. This results in the number in the far right column being output as a binary number in the three weld process selection bits shown on the Weld Output screen. These three bits select programs 1 to 8 on the Miller power supply. The weld process number 1 – 8 is specified in the DETAIL level of the weld schedule.

2.4 Selecting Miller Welding Modes

With the DeviceNet interface, ArcTool considers the Miller power supply to have 2 primary operating modes:

- **MIG** Which has 2 command signals:
 - o Voltage 0 – 100 Volts
 - o Wire feed speed 0 – 1400 IPM
- **Pulse** Which has 2 command signals:
 - o Arc Length (Trim) 0 – 100 (50 is nominal)
 - o Wire feed speed 0 – 1400 IPM

There isn't an automatic way to choose or switch between these two modes. Selection of the Miller mode can be accomplished by running a KAREL program called MI_EQ_PL.

There are 2 input parameters to the MI_EQ_PL program. The first parameter is the equipment number. The second parameter is an integer flag; 1 for Pulse and 0 for MIG.

MI_EQ_PL(1,0) = Equipment 1, MIG

MI_EQ_PL(1,1) = Equipment 1, PULSE

This KAREL program can be called in a weld program. You may need to set the system variable \$karel_enbl = 1 to be able to select the program name in the editor while editing the CALL instruction.

The following example program sets up the Miller interface for MIG mode.

```
TEST
1: !SET MIG
2: CALL MI_EQ_PL(1,0)
3:
4:J P[3] 40% FINE
: Arc Start E1[1]
5:L P[4] WELD_SPEED FINE
: Arc End E1[1]
[End]
```

The following example program sets up the Miller interface for Pulse mode.

```
TEST
1: !SET PULSE
2: CALL MI_EQ_PL(1,1)
3:
4:J P[3] 40% FINE
: Arc Start E1[1]
5:L P[4] WELD_SPEED FINE
: Arc End E1[1]
[End]
```

Running the MI_EQ_PL program sets the last 4 GO signals to appropriate values. These 4 signals indicate to the Miller power supply what command and feedback signals are desired. When Pulse mode is selected the GO values are set as shown below.

I/O	Group	Out	SIM	VALUE	1/100
GO[	1]	U	0	[E1 Weld Cmnd 1]	
GO[	2]	U	0	[E1 Weld Cmnd 2]	
GO[	3]	U	1	[E1 Req Param 1]	(Voltage feedback)
GO[	4]	U	0	[E1 Req Param 2]	(Current feedback)
GO[	5]	U	3	[E1 Sent Param 1]	(Arc Length command)
GO[	6]	U	5	[E1 Sent Param 2]	(Wire feed speed command)
GO[	7]	*	*	[	
GO[	8]	*	*	[	

Sorted by port number.
 [TYPE] CONFIG IN/OUT SIMULATE UNSIM >

When a program executes CALL MI_EQ_PL(1, 0) to select the Miller MIG mode, the Arc Length command is replaced by the Voltage command.

GO[	3]	U	1	[E1 Req Param 1]	(Voltage feedback)
GO[	4]	U	0	[E1 Req Param 2]	(Current feedback)
GO[	5]	U	4	[E1 Sent Param 1]	(Voltage command)
GO[	6]	U	5	[E1 Sent Param 2]	(Wire feed speed command)

It is useful to use the System Config menu to define a “COLD START Autoexec program” name. This TP program you specify will run on power up. It can contain a call of the MI_EQ_PL program.

2.5 Touch Sensing

If the touch sense option is installed on the robot controller the touch sense digital input and output signals must be assigned in the Touch Sense I/O Setup menu. The port numbers can be obtained from the digital input and output screens as shown below. This example is for a start point of 1. Your signal numbers may differ.

Touch I/O Setup 2/4

NAME	VALUE
1 Sensor port type:	DI
2 Sensor port number:	5
3 Circuit port type:	DO
4 Circuit port number:	12

[TYPE]
HELP

I/O Digital Out					1/512
#	SIM	STATUS			
DO[1]	U	OFF	[E1 Reserved		
DO[2]	U	OFF	[E1 Inch bwd		
DO[3]	U	OFF	[E1 Inch fwd		
DO[4]	U	OFF	[E1 Gas start		
DO[5]	U	OFF	[E1 Reserved		
DO[6]	U	OFF	[E1 Reserved		
DO[7]	U	OFF	[E1 Reserved		
DO[8]	U	OFF	[E1 Reserved		
DO[9]	U	OFF	[E1 Reserved		
DO[10]	U	OFF	[E1 Weld start		
DO[11]	U	OFF	[E1 Reserved		
DO[12]	U	OFF	[E1 Touch cmd		
DO[13]	U	OFF	[E1 Program 0		
DO[14]	U	OFF	[E1 Program 1		
DO[15]	U	OFF	[E1 Program 2		
DO[16]	U	OFF	[E1 Reserved		
DO[17]	U	OFF	[		
DO[18]	U	OFF	[		
DO[19]	U	OFF	[		
DO[20]	U	OFF	[		
DO[21]	U	OFF	[		
Sorted by port number.					
[TYPE]	CONFIG	IN/OUT	ON	OFF	>

I/O Digital In					1/512
#	SIM	STATUS			
DI[1]	U	OFF	[E1 Arc detect		
DI[2]	U	OFF	[E1 Bwd status		
DI[3]	U	OFF	[E1 Fwd status		
DI[4]	U	OFF	[E1 Gas status		
DI[5]	U	OFF	[E1 Touch fdbk		
DI[6]	U	OFF	[E1 Reserved		
DI[7]	U	OFF	[E1 Reserved		
DI[8]	U	OFF	[E1 Reserved		
DI[9]	U	OFF	[E1 Error		
DI[10]	U	OFF	[E1 Contactor		
DI[11]	U	OFF	[E1 Proc active		
DI[12]	U	OFF	[E1 Touch active		
DI[13]	U	OFF	[E1 Reserved		
DI[14]	U	OFF	[E1 Reserved		
DI[15]	U	OFF	[E1 Reserved		
DI[16]	U	OFF	[E1 Synergic		
DI[17]	U	OFF	[		
DI[18]	U	OFF	[		
DI[19]	U	OFF	[		
DI[20]	U	OFF	[		
DI[21]	U	OFF	[		
Sorted by port number.					
[TYPE]	CONFIG	IN/OUT	ON	OFF	>

3

ETHERNET

3 ETHERNET

3.1 Preliminary Steps

Before using the ArcTool setup menu at controlled start to select the Miller Auto-Axcess ENET interface you should perform these preliminary steps:

1. Clear Unused Ethernet I/O Assignments
2. Configure Multi-Process and Multi-Equipment
3. Plan the range of I/O port numbers used by the welder and other devices in the system.
4. Configure the Miller Weld Power Supply.

These preliminary steps are described in the next few sections. After these steps are completed the Miller Auto-Axcess ENET interface can be selected in the controlled start menu as described in Procedure 1-2.

3.1.1 Clear Unused Ethernet I/O Assignments

The Miller Ethernet interface requires the Ethernet I/O Scanner option R540. This option has a default setting that allocates 64 digital I/O signals to the first Ethernet connection as an Adaptor. This digital I/O and the Ethernet connection should be cleared if you do not plan to use it. This should be done before selecting the Miller Auto-Axcess ENET weld equipment interface. Use Procedure 1-1.

Procedure 3-1 Clearing the Default Ethernet I/O

Step

- 1 Press the MENU key and select I/O. Press TYPE to select Ethernet I/P. You should see a screen similar to the one below.

I/O Ethernet/IP					
Ethernet/IP List(Rack 89)					1/32
Description	TYP	Enable	Status	Slot	
Connection1	ADP	TRUE	OFFLINE	1	
Connection2	ADP	FALSE	OFFLINE	2	
[TYPE] PING CONFIG HELP					

- 2 Cursor to line 1, column 3. Change Connection1 Enable state from TRUE to FALSE.

- 3 Cursor to line 1, column 1. Press F4, CONFIG. The screen should be similar to the one shown below.

```

I/O Ethernet/IP
Adapter configuration :                      1/3

      Description :      Connection1
      Input size (words) : 4
      Output size(words) : 4
      Alarm Severity : WARN

      Scanner IP : *****
      API 0=>T :      0
      API T=>0 :      0

[ TYPE ]          PREV          HELP

```

- 4 Change the Input size and the Output size to 0 and then press F3, PREV.
- 5 Press the F1, TYPE and select Digital I/O. Press F2, CONFIG. The screen should be similar to the one shown below. The 64 signals associated with Rack 89 Slot 1 correspond to the default Connection1.

```

I/O Digital Out
                                     1/2
#      RANGE      RACK SLOT START STAT.
1 DO[   1- 64]    89   1    1 ACTIV
2 DO[  65- 512]    0   0    0 UNASG

[ TYPE ] MONITOR IN/OUT  DELETE  HELP

```

- 6 Clear the digital output assignments for Rack 89 Slot 1 by moving the cursor to this line and pressing the DELETE function key.
- 7 Press F3, IN/OUT to select the digital input screen.
- 8 Clear the digital input assignments for Rack 89 Slot 1 by moving the cursor to this line and pressing the DELETE function key.
- 9 Cycle power on the robot controller.

3.1.2 Configure Multi-Process and Multi-Equipment

The Miller ENET interface uses the ArcTool multi-process feature. Multiple process definitions allow you to select the various operating modes of the weld power supply. You should enable multi-process in the controlled start menu and cycle power before selecting the Miller ENET interface.

If you are connecting multiple weld equipment to the robot controller the multi-equipment option must be installed. You should set the correct number of equipment to be used in the controlled start setup menu and cycle power before selecting the Miller ENET interface.

Enabling multi-process and setting the number of weld equipment (and cycling power) can be done at the same time in the controlled start setup menu or they can be done as separate steps. These menu items are located near the bottom of the menu. From line 1 of the ArcTool controlled start setup menu you can cursor up to display these items.

```
ArcTool Setup
  8 Multi-process:           ENABLED
  9 Number of weld schedules: 32
 10 Number of weld equipment: 1
[ TYPE ]                     HELP
```

3.1.3 Plan the I/O range

If your robot system has other I/O devices it may be important to you where the welding I/O signals will be assigned within the overall I/O map.

The Miller ENET interface uses 80 digital inputs and 112 digital outputs. The first 16 inputs and outputs are individual control signals. The remaining signals are configured as 7 group outputs and 4 group inputs.

During the controlled start setup procedure you have the opportunity to enter a start point for the Digital I/O signals and for the Group I/O signals. You can also simply allow the signal numbering to begin at the first available area.

Start points begin one after a multiple of 8 (1, 9, 17, etc.) The number you enter will be used to search for an available I/O range. The full range of contiguous signals (80 inputs and 112 outputs) must be available without conflict with a previous assignment.

3.1.4 Configure the Miller Weld Power Supply

To complete the setup on the robot you will need to know the IP address to be used by the Miller weld equipment.

The Miller Auto-Axcess ENET power supply has 2 Ethernet connections. The connectors are located on the back of the machine. They are labeled A and B. Port B should be connected to Ethernet Port 2 on the robot controller. Port A can be connected to a PC to provide access to the Miller machine's local web page.

The IP address of Port B on the Miller Auto-Axcess ENET power supply can be found using the Miller machine's web pages. Connect a PC to Port A with an Ethernet cable. Create a Local Area Network connection and login to the Miller machine's home page with a web browser. The default factory setting for the IP address of Port A is 169.254.0.2. The subnet mask is 255.255.0.0.

From the Miller home page navigate to System Settings to find the Ethernet setup.

The power supply's Port B IP address must be on the robot port 2 subnet. Typically, Ethernet port 2 of the robot is given the IP address 192.168.2.161 and the Miller weld power supplies are assigned IP addresses in sequence, as needed.

192.168.2.161 (Robot Port 2)
192.168.2.162 (Equipment 1)
192.168.2.163 (Equipment 2)
192.168.2.164 (Equipment 3)
192.168.2.165 (Equipment 4)

The Miller Auto-Axcess ENET has many setup items and can be configured to communicate using either Ethernet or DeviceNet. The menu for this information is located under Robot Configuration / Robot Options Setup. In this menu you should verify:

- The Robot Communication Method is Ethernet IP.
- The Remote Program Select feature is enabled.
- Robot Control is selected.
- The Robot Manufacturer is FANUC.

The slow and fast wire jog speeds are setup on the Miller Welder Config / Feed Torch web page. Note that these speeds are used for wire inching, not the similar items listed on the robot's weld equipment setup screen.

3.2 Weld Power Supply Selection, Setup, and Verification

After completing the preliminary steps of the previous section, the weld power supply setup begins by selecting the interface in the controlled start setup menu.

3.2.1 Weld Power Supply Selection

Use the following procedure to configure ArcTool for operation with a Miller Auto-Axcess ENET weld power supply.

Procedure 3-2 Selecting a Miller Weld Power Supply at Controlled Start Conditions

- The Ethernet I/O Scanner option R540 is installed.
- The Miller Weld Equipment Library Option R580 is installed.
- You have completed the Preliminary Steps in section 1.2.
- The ArcTool Setup menu is displayed at the Controlled Start as shown below.

ArcTool Setup		6/10
1 F Number:	F00000	
2 Welding setup:	USA	
3 Wire speed units:	IPM	
4 Weld speed units:	IPM	
5 Weld speed:	50	
6 Manufacturer:	General Purpose	
7 Model:	MIG (Volts, WFS)	
Press FCTN then START (COLD) when done.		
[TYPE]	CHECK	[CHOICE] HELP

Step

- 1 Move the cursor to the Manufacturer field and press F4, [CHOICE]. Select Miller. The Model name will default to Deltaweld452.
- 2 Move the cursor the Model field and press F4, [CHOICE] and select Auto-Axcess ENET from the list.
- 3 You will asked if you would like to specify a starting digital I/O number. If you choose not to specify a number, the I/O will be located at the first block of adequate size. If you specify a starting number the I/O will begin at or after this point. You will be asked the same question regarding the starting number for the group I/O.
- 4 Press FCTN, Select START(Cold), and press ENTER to Cold Start the controller.

After choosing the Miller weld power supply model at controlled start there are a few additional setup steps to perform.

3.2.2 Configure the Robot IP Address

You must configure the IP address of Ethernet port 2 on the robot. The welding equipment should be on a separate network from the system or plant level interface. Note that Ethernet ports 1 and 2 in the robot controller must be mapped to different subnets. Port 2 is typically reserved for communication to the welding equipment.

Procedure 3-3 Configuring the Robot IP Address

Step

- 1** Press MENU, SETUP, HOST COM.

SETUP Protocols		1/7
Protocol	Description	
1 TCP/IP	TCP/IP Detailed Setup	
2 TELNET	Telnet Protocol	
3 PPP	Point to Point Protocol	
4 PING	Ping Protocol	
5 HTTP	HTTP Authentication	
6 FTP	File Transfer Protocol	
7 SMTP	EMAIL Setup	
[TYPE]		DETAIL [SHOW]

- 2** On line 1, press F3, DETAIL

SETUP Host Comm		1/40
TCP/IP		
Robot name:	ROBOT	
Port#1 IP addr:	*****	
Subnet Mask:	255.255.255.0	
Board address:	*****	
Router IP addr:	*****	
Host Name (LOCAL) Internet Address		
1 *****	*****	
2 *****	*****	
3 *****	*****	
4 *****	*****	
[TYPE]		PORT PING HELP >

- 3** Press F3, PORT so line 2 displays Port #2 IP addr.

Port#2 IP addr:*****

- 4** Cursor to Port #2 IP addr: and Press ENTER.
5 Enter the IP address: 192.168.2.161
6 Verify there are no spaces in the IP address.

TCP/IP	2/40
Robot name:	ROBOT
Port#2 IP addr:	192.168.2.161

3.2.3 Configure the Ethernet Scanner with Power Supply IP Address

The IP address assigned to the weld power supply must be entered in the robot Ethernet I/O setup screen.

Procedure 3-4 Configuring the Ethernet Connection IP Address

Step

- 1 Press the MENU key and select I/O. Press F1, TYPE and select Ethernet I/P. You should see a screen similar to the one below.

I/O Ethernet/IP					
Ethernet/IP List(Rack 89)					1/32
Description	TYP	Enable	Status	Slot	
Miller_EQ1	SCN	TRUE	OFFLINE	1	
Connection2	ADP	FALSE	OFFLINE	2	
Connection3	ADP	FALSE	OFFLINE	3	
Connection4	ADP	FALSE	OFFLINE	4	
Connection5	ADP	FALSE	OFFLINE	5	
[TYPE] PING CONFIG HELP					

- 2 Cursor to line 1, column 3. Change Miller_EQ1 Enable to FALSE.
- 3 Cursor to line 1, column 1. Press F4, CONFIG. The screen should be similar to the one below.

I/O Ethernet/IP	
Scanner configuration :	1/10
Description :	Miller_EQ1
Name/IP address :	*****
Vendor Id :	486
Device Type :	43
Product code :	1
Input size (bytes):	10
Output size (bytes):	14
RPI (ms) :	10
Assembly instance(input) :	100
Assembly instance(output) :	112
Configuration instance :	1
[TYPE] ADV PREV ANALOG HELP	

- 4 Cursor to Name/IP address: and Press ENTER.
- 5 Enter the IP address: 192.168.2.162 for Equipment 1.
- 6 Verify there are no spaces in the IP address.
- 7 Press PREV.
- 8 Cursor to line 1, column 3. Change Miller_EQ1 Enable to TRUE.
- 9 Cycle power on the robot controller and the Miller weld power supply.

3.2.4 Verify Scanner Operation

After power up, revisit the Ethernet I/P screen to confirm the robot and weld power supply are communicating.

Press the MENU key and select I/O. Press F1, TYPE and select Ethernet I/P. You should see a screen similar to the one below.

I/O Ethernet/IP					
Ethernet/IP List(Rack 89)				1/32	
Description	TYP	Enable	Status	Slot	
Miller_EQ1	SCN	TRUE	RUNNING	1	
Connection2	ADP	FALSE	OFFLINE	2	
Connection3	ADP	FALSE	OFFLINE	3	
[TYPE] PING CONFIG HELP					

The scanner status should indicate "RUNNING".

3.2.5 Verify the Digital I/O Assignments

The digital I/O signals were automatically configured during the controlled start setup. Comments were added to the 16 digital control signals, but not to those used as group I/O. Some of the signals in the Miller I/O range are not used.

I/O Digital Out					
	#	SIM	STATUS	1/512	
DO[	1]	U	OFF	[E1 Reserved	]
DO[	2]	U	OFF	[E1 Inch bwd	]
DO[	3]	U	OFF	[E1 Inch fwd	]
DO[	4]	U	OFF	[E1 Gas start	]
DO[	5]	U	OFF	[E1 Reserved	]
DO[	6]	U	OFF	[E1 Reserved	]
DO[	7]	U	OFF	[E1 Reserved	]
DO[	8]	U	OFF	[E1 Reserved	]
DO[	9]	U	OFF	[E1 Reserved	]
DO[	10]	U	OFF	[E1 Weld start	]
DO[	11]	U	OFF	[E1 Reserved	]
DO[	12]	U	OFF	[E1 Touch cmd	]
DO[	13]	U	OFF	[E1 Reserved	]
DO[	14]	U	OFF	[E1 Reserved	]
DO[	15]	U	OFF	[E1 Reserved	]
DO[	16]	U	OFF	[E1 Reserved	]
DO[	17]	U	OFF	[	]
DO[	18]	U	OFF	[	]
DO[	19]	U	OFF	[	]
DO[	20]	U	OFF	[	]
DO[	21]	U	OFF	[	]
Sorted by port number.					
[TYPE]	CONFIG	IN/OUT	ON	OFF	>

I/O Digital In					
	#	SIM	STATUS	1/512	
DI[	1]	U	OFF	[E1 Arc detect	]
DI[	2]	U	OFF	[E1 Bwd status	]
DI[	3]	U	OFF	[E1 Fwd status	]
DI[	4]	U	OFF	[E1 Gas status	]
DI[	5]	U	OFF	[E1 Touch fdbk	]
DI[	6]	U	OFF	[E1 Reserved	]
DI[	7]	U	OFF	[E1 Reserved	]
DI[	8]	U	OFF	[E1 Reserved	]
DI[	9]	U	OFF	[E1 Error	]
DI[	10]	U	OFF	[E1 Contactor	]
DI[	11]	U	OFF	[E1 Proc active	]
DI[	12]	U	OFF	[E1 Touch active	]
DI[	13]	U	OFF	[E1 Reserved	]
DI[	14]	U	OFF	[E1 Reserved	]
DI[	15]	U	OFF	[E1 Reserved	]
DI[	16]	U	OFF	[E1 Synergic	]
DI[	17]	U	OFF	[	]
DI[	18]	U	OFF	[	]
DI[	19]	U	OFF	[	]
DI[	20]	U	OFF	[	]
DI[	21]	U	OFF	[	]
Sorted by port number.					
[TYPE]	CONFIG	IN/OUT	ON	OFF	>

The CONFIG function key can be used to confirm the correct number of signals are mapped (112 outputs and 80 inputs).

I/O Digital Out						I/O Digital In					
#	RANGE	RACK	SLOT	START	STAT.	#	RANGE	RACK	SLOT	START	STAT.
1	DO[1- 112]	89	1	1	ACTIV	1	DI[1- 80]	89	1	1	ACTIV
2	DO[113- 512]	0	0	0	UNASG	2	DI[81- 512]	0	0	0	UNASG
Device Name : EthernetIP						Device Name : EthernetIP					
[TYPE] MONITOR IN/OUT DELETE HELP						[TYPE] MONITOR IN/OUT DELETE HELP					

Pressing the SHIFT-Wire +/- hard keys feeds the filler wire. This can be used to help verify the digital output operation.

3.2.6 Verify the Group I/O Assignments

The group inputs and outputs signal names should match those shown in the following figure. The port numbers shown below are based on a starting port number of 1.

I/O Group Out						I/O Group In					
#	SIM	VALUE			1/100	#	SIM	VALUE			1/100
GO[1]	U	325	[E1 Wire feed		]	GI[1]	U	0	[E1 Wire feed		]
GO[2]	U	500	[E1 Weld Cmnd 2		]	GI[2]	U	0	[E1 Current		]
GO[3]	U	25	[E1 Weld Cmnd 3		]	GI[3]	U	0	[E1 Voltage		]
GO[4]	U	30	[E1 Weld Cmnd 4		]	GI[4]	U	0	[E1 Error		]
GO[5]	U	1	[E1 Program		]	GI[5]	*	*	[		]
GO[6]	U	0	[E1 Part ID		]	GI[6]	*	*	[		]
GO[7]	U	0	[E1 Weld ID		]	GI[7]	*	*	[		]
GO[8]	*	*	[		]	GI[8]	*	*	[		]
GO[9]	*	*	[		]	GI[9]	*	*	[		]
GO[10]	*	*	[		]	GI[10]	*	*	[		]
Sorted by port number.						Sorted by port number.					
[TYPE] CONFIG IN/OUT SIMULATE UNSIM >						[TYPE] CONFIG IN/OUT SIMULATE UNSIM >					

The CONFIG function key can be used to show how many signals (NUM PTS) are mapped to each port.

I/O Group Out						I/O Group In					
GO #	RACK	SLOT	START PT	NUM	PTS	GI #	RACK	SLOT	START PT	NUM	PTS
1	89	1	17	16		1	89	1	17	16	
2	89	1	33	16		2	89	1	33	16	
3	89	1	49	8		3	89	1	49	16	
4	89	1	57	8		4	89	1	65	16	
5	89	1	65	16		5	0	0	0	0	
6	89	1	81	16		6	0	0	0	0	
7	89	1	97	16		7	0	0	0	0	
8	0	0	0	0		8	0	0	0	0	
[TYPE] MONITOR IN/OUT HELP >						[TYPE] MONITOR IN/OUT HELP >					

3.2.7 System Variable Settings

Choosing the Miller Ethernet interface during controlled start disables burnback control by the robot and enables 2 Weld End features. This allows burnback to be controlled by the Miller weld power supply and requires the robot to wait for the machine to complete welding.

\$awspcr.\$bback_enabl = FALSE The burnback weld schedule values and time delay are not used during the Weld End sequence.

\$awepr[eq].\$ae_hold_ao = TRUE The weld command signals (analog or group outputs) are left at their last values when the Weld End completes.

\$awepr[eq].\$ae_chk_di = TRUE If \$ae_chk_di is TRUE the execution of an Weld End will delay after burnback until the Process Active input signal is OFF before proceeding with postflow and wirestick checking. The maximum delay before an error is reported is controlled by \$aweupr[eq].\$max_end_time

\$aweioex[eq].\$proc_active.\$port_num This port number is assigned to the Process Active signal of the Miller interface. The Weld End will wait for this signal to be OFF. If the time exceeds the max_end_time the error "ARC-250 Process active ON after weld" is reported.

3.3 Weld Procedure Data Screen

The weld procedures listed in the DATA screen may not be configured properly for the Miller machine if you had another equipment configuration. If you delete all the weld procedures in the Weld Procedure DATA screen and cycle power, the system will create 2 new weld procedures during power up. Procedure 1 will be configured for MIG welding and Procedure 2 will be configured for Synergic welding. These are the two primary process modes which can be used with the Miller machine.

DATA Weld Procedure 1		1/6
+ Procedure	1	[]
+ Mode	[MIG	]
+ Schedules		

+ Procedure	2	[]
+ Mode	[Synergic	]
+ Schedules		
[TYPE] DETAIL [CMND][VIEW] HELP		

When a MIG procedure is selected, there are 4 welding command signals:

- Wire Feed Speed 0 – 1400 IPM
- Voltage 0 – 44 Volts
- Inductance 0 – 99 (50 is nominal)
- Slope 0 – 99 (50 is nominal)

The Miller power supply may have multiple Synergic modes, such as AccuPulse and RMD. Miller modes that are not named “MIG” should all be programmed as “Synergic” on the robot. These modes use these 3 welding command signals:

- Wire Feed Speed 0 – 1400 IPM
- Arc Length 0 – 99 (50 is nominal)
- Arc Sharp 0 – 50 (25 is nominal)

If you cursor to the mode description (MIG/Synergic) and press function key F4, CHOICE, the mode can be changed. The Mode specified in a procedure (MIG or Synergic) must correspond to the mode used in the specified program on the Miller welder.

The Miller power supply can have up to 8 programs defined (the maximum number may increase). Each of these programs can be configured on the Miller front panel to use a particular weld mode (MIG, AccuPulse, RMD, etc.).

There is a program number defined in each weld procedure on the robot controller. This program number corresponds to the program number on the Miller power supply. The program number is sent to the welder as a group output for the currently selected weld procedure. If you cursor to the "+" in front of the Mode and press the ENTER key, the program number will be displayed and can be modified.

```

DATA Weld Procedure  1
                                     3/7
+ Procedure          1 [           ]
- Mode               [MIG          ]
  Program:           1
+ Schedules

-----

+ Procedure          2 [           ]
- Mode               [Synergic     ]
  Program:           2
+ Schedules

[ TYPE ] DETAIL [ CMND ][ VIEW ]  HELP

```

You can manually select a weld procedure by positioning the cursor on a weld procedure and choosing the SELECT key listed under the CMND function key. When you select a MIG procedure the Weld I/O, Weld Status, and the OnTheFly screen will contain the 4 welding command signals shown below.

```

I/O Weld Out
                                     1/12
      WELD SIGNAL      TYPE #  SIM STATUS
1 [Wire feed          ] GO[  1] U   0.0
2 [Voltage             ] GO[  2] U   0.0
3 [Inductance          ] GO[  3] U   0.0
4 [Slope               ] GO[  4] U   0.0

5 [Weld start          ] DO[ 10] U  OFF
6 [Gas start           ] DO[  4] U  OFF
7 [Inch forward        ] DO[  3] U  OFF
8 [Inch backward       ] DO[  2] U  OFF
9 [Program number      ] GO[  5] U    1

10 [Wire stick alarm]  [****] *  ***
11 [Proc select 1     ] [****] *  ***
12 [Proc select 3     ] [****] *  ***

[ TYPE ]  HELP  IN/OUT SIMULATE UNSIM >

```

If you select a Synergic weld procedure the screens are modified to display the 3 welding command signals shown below.

I/O Weld Out					
					1/11
	WELD SIGNAL	TYPE #	SIM	STATUS	
1	[Wire feed	] GO[1]	U	0.0	
2	[Arc Length	] GO[2]	U	0.0	
3	[Arc Sharp	] GO[3]	U	0.0	
4	[Weld start	] DO[10]	U	OFF	
5	[Gas start	] DO[4]	U	OFF	
6	[Inch forward	] DO[3]	U	OFF	
7	[Inch backward	] DO[2]	U	OFF	
8	[Program number	] GO[5]	U	2	
9	[Wire stick alarm]	[****]	*	***	
10	[Proc select 1	] [****]	*	***	
11	[Proc select 3	] [****]	*	***	
[TYPE] HELP IN/OUT SIMULATE UNSIM >					

Note the SELECT command caused the program number GO value on line 8 to change from 1 to 2. This selects program 2 on the Miller welder.

3.4 Touch Sensing

If the touch sense option is installed on the robot controller the touch sense digital input and output signals must be assigned in the Touch Sense I/O Setup menu. The port numbers can be obtained from the digital input and output screens as shown below. This example is for a start point of 1. Your signal numbers may differ.

Touch I/O Setup		2/4
NAME	VALUE	
1 Sensor port type:	DI	
2 Sensor port number:	5	
3 Circuit port type:	DO	
4 Circuit port number:	12	
[TYPE]		HELP

I/O Digital Out					1/512
#	SIM	STATUS			
DO[1]	U	OFF	[E1 Reserved		
DO[2]	U	OFF	[E1 Inch bwd		
DO[3]	U	OFF	[E1 Inch fwd		
DO[4]	U	OFF	[E1 Gas start		
DO[5]	U	OFF	[E1 Reserved		
DO[6]	U	OFF	[E1 Reserved		
DO[7]	U	OFF	[E1 Reserved		
DO[8]	U	OFF	[E1 Reserved		
DO[9]	U	OFF	[E1 Reserved		
DO[10]	U	OFF	[E1 Weld start		
DO[11]	U	OFF	[E1 Reserved		
DO[12]	U	OFF	[E1 Touch cmd		
DO[13]	U	OFF	[E1 Program 0		
DO[14]	U	OFF	[E1 Program 1		
DO[15]	U	OFF	[E1 Program 2		
DO[16]	U	OFF	[E1 Reserved		
DO[17]	U	OFF	[		
DO[18]	U	OFF	[		
DO[19]	U	OFF	[		
DO[20]	U	OFF	[		
DO[21]	U	OFF	[		
Sorted by port number.					
[TYPE] CONFIG IN/OUT ON OFF >					

I/O Digital In					1/512
#	SIM	STATUS			
DI[1]	U	OFF	[E1 Arc detect		
DI[2]	U	OFF	[E1 Bwd status		
DI[3]	U	OFF	[E1 Fwd status		
DI[4]	U	OFF	[E1 Gas status		
DI[5]	U	OFF	[E1 Touch fdbk		
DI[6]	U	OFF	[E1 Reserved		
DI[7]	U	OFF	[E1 Reserved		
DI[8]	U	OFF	[E1 Reserved		
DI[9]	U	OFF	[E1 Error		
DI[10]	U	OFF	[E1 Contactor		
DI[11]	U	OFF	[E1 Proc active		
DI[12]	U	OFF	[E1 Touch active		
DI[13]	U	OFF	[E1 Reserved		
DI[14]	U	OFF	[E1 Reserved		
DI[15]	U	OFF	[E1 Reserved		
DI[16]	U	OFF	[E1 Synergic		
DI[17]	U	OFF	[		
DI[18]	U	OFF	[		
DI[19]	U	OFF	[		
DI[20]	U	OFF	[		
DI[21]	U	OFF	[		
Sorted by port number.					
[TYPE] CONFIG IN/OUT ON OFF >					

4

DELTAWELD 452

4 DELTAWELD 452

Procedure 4-1 Setup Procedure for the Deltaweld 452

Step

- 1 Perform a controlled start.
 - a If the controller is on, turn it off.
 - b On the teach pendant, press and hold the PREV and NEXT keys.
 - c While still pressing PREV and NEXT on the teach pendant, turn the power disconnect circuit breaker to ON. You will see a screen similar to the following.

```
----- CONFIGURATION MENU -----  
-  
1 Hot start  
2 Cold start  
3 Controlled start  
4 Maintenance
```

- d After the configuration menu appears on the screen, release the PREV and NEXT keys.
 - e Select 3, Controlled start.
- 2 In the ArcTool Setup menu cursor to the **Multi-process** field and select DISABLED. The Miller Deltaweld 452 interface does not require this feature.
- 3 If you are using multiple weld equipment, cursor to the **Number of weld equipment** field and enter the correct number. This field cannot be set if the multi-equipment option is not installed.

```
ArcTool Setup  
  
8 Multi-process:          DISABLED  
9 Weld ID:                DISABLED  
10 Number of weld schedules: 32  
11 Number of weld equipment: 1  
  
[ TYPE ]                  HELP
```

- 4 Beginning with version 8.10 ArcTool uses Weld Procedures as the default programming method. Weld Procedures use Weld Start and Weld End instructions instead of Arc Start and Arc End. This feature must be disabled to use the Miller **Deltaweld 452** interface. Weld Procedures are used with the Miller **Ethernet** interface.
 - a Press the MENU key and select Variables.
 - b Cursor to the system variable \$AWSCFG and press ENTER.
 - c Cursor to the field \$WLD_PRC_ENA and set it to FALSE.
- 5 Turn the controller off and on. The ArcTool setup menu should be displayed.

ArcTool Setup		6/10
1 F Number:	F00000	
2 Welding setup:	USA	
3 Wire speed units:	IPM	
4 Weld speed units:	IPM	
5 Weld speed:	50	
6 Manufacturer:	General Purpose	
7 Model:	MIG (Volts, WFS)	
Press FCTN then START (COLD) when done.		
[TYPE]	CHECK	[CHOICE] HELP

6 Move the cursor to the Manufacturer field and press F4, [CHOICE]. Select Miller. The Model name will default to Deltaweld452.

7 Press FCTN, Select START(Cold), and press ENTER to Cold Start the controller.

The welding I/O signals should be mapped as shown below.

I/O Weld Out					I/O Weld In				
				1/11					1/12
WELD SIGNAL	TYPE #	SIM	STATUS		WELD SIGNAL	TYPE #	SIM	STATUS	
1 [Voltage	] AO[	1]	U	0.0	1 [Voltage	] AI[	1]	U	0.0
2 [Wire feed	] AO[	2]	U	0.0	2 [Current	] AI[	2]	U	0.0
3 [Weld start	] WO[	1]	U	OFF	3 [	] WI[	1]	U	OFF
4 [Gas start	] WO[	2]	U	OFF	4 [Arc detect	] WI[	2]	U	OFF
5 [	] WO[	3]	U	OFF	5 [Gas fault	] WI[	3]	U	OFF
6 [Inch forward	] WO[	4]	U	OFF	6 [Wire fault	] WI[	4]	U	OFF
7 [Inch backward	] WO[	5]	U	OFF	7 [Water fault	] WI[	5]	U	OFF
8 [Proc select 1	] WO[	6]	U	OFF	8 [Power fault	] WI[	6]	U	OFF
9 [Proc select 2	] WO[	7]	U	OFF	9 [	] WI[	7]	U	OFF
10 [Proc select 3	] WO[	8]	U	OFF	10 [	] WI[	8]	U	OFF
11 [Wire stick alarm]	[****]	*	***		11 [Wirestick	] WS[	1]	U	OFF
					12 [Arc enable	] [****]	*	***	
[TYPE]	HELP	IN/OUT	SIMULATE	UNSIM >	[TYPE]	HELP	IN/OUT	SIMULATE	UNSIM >

The 3 Process Select bits are not used by this power supply. This feature can be disabled by setting \$awepcr[1].\$prg_sel_ena = FALSE.

5

AUTO-INVISION

5 AUTO-INVISION

Procedure 5-1 Setup Procedure for the Auto-Invision

Step

- 1 Perform a controlled start.
 - a If the controller is on, turn it off.
 - b On the teach pendant, press and hold the PREV and NEXT keys.
 - c While still pressing PREV and NEXT on the teach pendant, turn the power disconnect circuit breaker to ON. You will see a screen similar to the following.

```
----- CONFIGURATION MENU -----  
-  
1 Hot start  
2 Cold start  
3 Controlled start  
4 Maintenance
```

- d After the configuration menu appears on the screen, release the PREV and NEXT keys.
 - e Select 3, Controlled start.
- 2 In the ArcTool Setup menu cursor to the **Multi-process** field and select DISABLED. The Miller Deltaweld 452 interface does not require this feature.
- 3 If you are using multiple weld equipment, cursor to the **Number of weld equipment** field and enter the correct number. This field cannot be set if the multi-equipment option is not installed.

```
ArcTool Setup  
  
8 Multi-process:          DISABLED  
9 Weld ID:                DISABLED  
10 Number of weld schedules: 32  
11 Number of weld equipment: 1  
  
[ TYPE ]                  HELP
```

- 4 Beginning with version 8.10 ArcTool uses Weld Procedures as the default programming method. Weld Procedures use Weld Start and Weld End instructions instead of Arc Start and Arc End. This feature must be disabled to use the Miller **Auto-Invision** interface. Weld Procedures are used with the Miller **Ethernet** interface.
 - a Press the MENU key and select Variables.
 - b Cursor to the system variable \$AWSCFG and press ENTER.
 - c Cursor to the field \$WLD_PRC_ENA and set it to FALSE.

- 5 Turn the controller off and on. The ArcTool setup menu should be displayed.

ArcTool Setup		6/10
1 F Number:	F00000	
2 Welding setup:	USA	
3 Wire speed units:	IPM	
4 Weld speed units:	IPM	
5 Weld speed:	50	
6 Manufacturer:	General Purpose	
7 Model:	MIG (Volts, WFS)	
Press FCTN then START (COLD) when done.		
[TYPE]	CHECK	[CHOICE] HELP

- 6 Move the cursor to the Manufacturer field and press F4, [CHOICE]. Select Miller. The Model name will default to Deltaweld452.
- 7 Move the cursor the Model field and press F4, [CHOICE] and select Auto-Axcess DNET from the list.
- 8 Press FCTN, Select START(Cold), and press ENTER to Cold Start the controller.

The welding I/O signals should be mapped as shown below.

I/O Weld Out					I/O Weld In				
				1/11					1/12
WELD SIGNAL	TYPE #	SIM	STATUS		WELD SIGNAL	TYPE #	SIM	STATUS	
1 [Voltage	] AO[	1] U	0.0		1 [Voltage	] AI[	1] U	0.0	
2 [Wire feed	] AO[	2] U	0.0		2 [Current	] AI[	2] U	0.0	
3 [Weld start	] WO[	1] U	OFF		3 [	] WI[	1] U	OFF	
4 [Gas start	] WO[	2] U	OFF		4 [Arc detect	] WI[	2] U	OFF	
5 [	] WO[	3] U	OFF		5 [Gas fault	] WI[	3] U	OFF	
6 [Inch forward	] WO[	4] U	OFF		6 [Wire fault	] WI[	4] U	OFF	
7 [Inch backward	] WO[	5] U	OFF		7 [Water fault	] WI[	5] U	OFF	
8 [Proc select 1	] WO[	6] U	OFF		8 [Power fault	] WI[	6] U	OFF	
9 [Proc select 2	] WO[	7] U	OFF		9 [	] WI[	7] U	OFF	
10 [Proc select 3	] WO[	8] U	OFF		10 [	] WI[	8] U	OFF	
11 [Wire stick alarm]	[****]	*	***		11 [Wirestick	] WS[	1] U	OFF	
					12 [Arc enable	] [****]	*	***	
[TYPE]	HELP	IN/OUT	SIMULATE	UNSIM >	[TYPE]	HELP	IN/OUT	SIMULATE	UNSIM >